



3M at a glance

Source: 3M annual report 2017

Sales in 200 countries

31.7 billion in sales

Five Business groups

91,000 3Mers globally

113,000 patents

101 straight years of dividends

Case Study Overview



Background

Facts / Variables

Objective / Challenge

Volume File

Solution / Results

A 2020 Vision for US Supply Chain Redesign

Background

While 3M customer expectations *for faster shipping with shorter cycle times continued to rise, 3M distribution structure was feeling the strain, so company executives set out to complete a supply chain redesign to meet customer requirements even while those requirements were sure to evolve.*

Supply chain network design was critical to realizing step-change improvements in service. At the same time, 3M needed to continue driving down cost-to-serve and continue improving the quality of shipments.

Graham Stowe, Vice President, Planning & Logistics

at 3M closed his office door. In his last day of data review before creating a 3M 2020 Vision for US Supply Chain redesign, he didn't want any interruptions. The redesign would shape how 3M would respond to changes in fulfillment and shipment that had reached a tipping point for 3M. Annual incremental patches had made improvements to the 3M supply chain, but in the age of Amazon, now was the time for a significant redesign for sustainable growth and a leap forward in 3M supply chain operations.

A well-thumbed copy of *Supply Chain Management Review* was open on his desk where one passage highlighted the one force that had emerged as a powerful factor even for businesses as large and well-established as 3M:

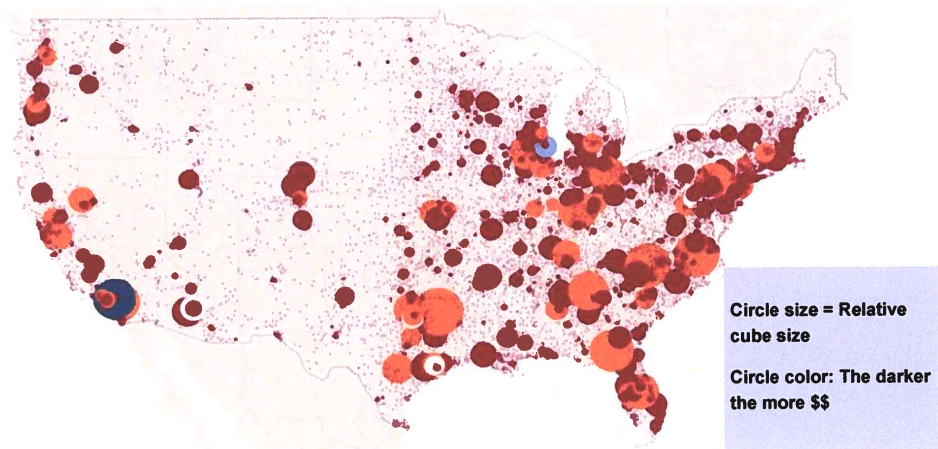
"...in supply chain operations, the Amazon Effect is causing a revolution of epic proportions across the globe."
(Coates, 2017).

Fast, accurate shipping. For 3M, that meant one day for order processing and one day for shipping to the final destination. Epic.



Background

The map on Stowe's screen told the tale of the incremental growth that made sense before 2-day delivery became as common as drive-through coffee.



Customers wanted a higher level of service in a shorter cycle time. 3M had the resources for inventory availability confirmation, shipment confirmation, and delivery notification. What 3M didn't have was a plan for 2-day shipping. Graham printed out the big question and slapped it on the wall:

How should 3M redesign its supply chain network for faster shipping with shorter cycle times?

We need to have the right product in the right place at the right time to fulfill demand. We need a distribution network strategy that allows us to meet requirements.



Background

One thing was clear: simple answers never worked

“Let’s boost inventories in all our DCs!” Too expensive and way too inefficient.

“Let’s build dozens of regional DCs!” While labor generally wasn’t a factor, new DCs were a solid \$1 million apiece in annual administration costs. What would it take in inventory management, transportation cost and speed, and DC design to justify the cost of each DC?

“Technology to the rescue!” Maybe, but which technology to achieve which goals, at what scale and at what cost? 3M would not be flying drones out of DCs for deliveries. It was one thing to run down a list of technology solutions but each one had to be balanced by the realities on the ground. And technology didn’t control the weather. The fact was, in some locations they were only a snowstorm or heavy wind away from the next transportation problem, power or internet outage or both.

Weather notwithstanding, how could the 3M enterprise resource planning (ERP) platform and integrated business planning (IBP) process optimize their supply chain for meeting the 2-day delivery goal? 3M had to get close to customers, but profitably. Right now 3M had too much variability in a transportation network that could meet only 45% of customer demand for 2-day shipping.

1Day												
	A	B	C	D	E	F	G	H	I	J	K	L
1	CUST_NAME	Orders	Lines	OrderBU	STATE_ABBR	TotalCartons	TotalCube	TotalWeight	ZIP_CODE	PartNum	ItemActivityCode	PickPa
2	CUSTOMER A	33	33	9632	FL	9632	2465.792	21556.416	349814620	PART7023305	A	1Day
3	CUSTOMER A	30	30	8064	FL	8064	2064.384	18047.232	346028310	PART7023305	A	1Day
4	CUSTOMER A	28	28	7840	PA	7840	2007.04	17545.92	184668990	PART7023305	A	1Day
5	CUSTOMER A	29	29	6944	NH	6944	1777.664	15540.672	30772385	PART7023305	A	1Day
6	CUSTOMER A	20	20	6720	IL	6720	1720.32	15039.36	613629133	PART7023305	A	1Day
7	CUSTOMER A	26	26	6048	NY	6048	1548.288	13535.424	134033030	PART7023305	A	1Day
8	CUSTOMER A	23	23	5824	CO	5824	1490.944	13034.112	805388958	PART7023305	A	1Day
9	CUSTOMER A	23	23	5600	TX	5600	1433.6	12532.8	762663547	PART7023305	A	1Day
10	CUSTOMER D	10	10	22192	PA	5548	4077.78	13448.352	170269321	PART6963196	A	1Day
11	CUSTOMER A	23	23	5376	TX	5376	1376.256	12031.488	790721951	PART7023305	A	1Day
12	CUSTOMER A	20	20	5376	TX	5376	1376.256	12031.488	774749516	PART7023305	A	1Day
13	CUSTOMER A	17	17	5355	FL	5355	717.57	3496.815	346028310	PART5286391	A	1Day
14	CUSTOMER A	16	16	5040	FL	5040	675.36	3291.12	349814620	PART5286391	A	1Day

“More trucks!” Again—maybe. That solution always depended on time, distance, average speeds, and of course, inventory management and demand forecasting. 3M had plenty of shipping data on its thousands of shipments each quarter.



As little as two years ago, the Amazon effect wasn't yet a significant factor in supply chain design. Now 3M had to improve business processes for placing the right people and 65,000 products in the right location at the right time.

How could the 3M enterprise resource planning (ERP) platform and integrated business planning (IBP) process optimize their supply chain for meeting the 2-day delivery goal? 3M had to get close to customers, but profitably. Right now 3M had too much variability in a transportation network that could meet only 45% of customer demand for 1-day fulfillment and 1-day delivery.

Agile logistics at 3M

The company couldn't expect to meet increasing service expectations using the same data and planning processes or manufacturing and distribution strategy as they had in the past. 3M prided itself on innovation and agility--so what would a more agile supply chain look like? In spite of popular misconceptions, agility didn't mean being able to jump over boxes. It meant providing timely, accurate customer information to everyone who needed it to be productive; it meant frequent review of actions to assess their outcomes; and it meant adapting to what you learned along the way.

An agile technique they'd used to meet specific customer demands in the past was to create user stories about functionality customers wanted and how each functionality met specific needs. Graham added three to the whiteboard, but this time with a focus on 3M demands.

As a transportation manager, I want to keep my trucks full so I can maximize value per trip.

As an executive responsible for managing expenses, I want to know that technology investments will result in a measurable return so I can justify the spending.

As a fulfillment manager, I want accurate real-time demand data so I can manage the flow of goods in and out of my DC.

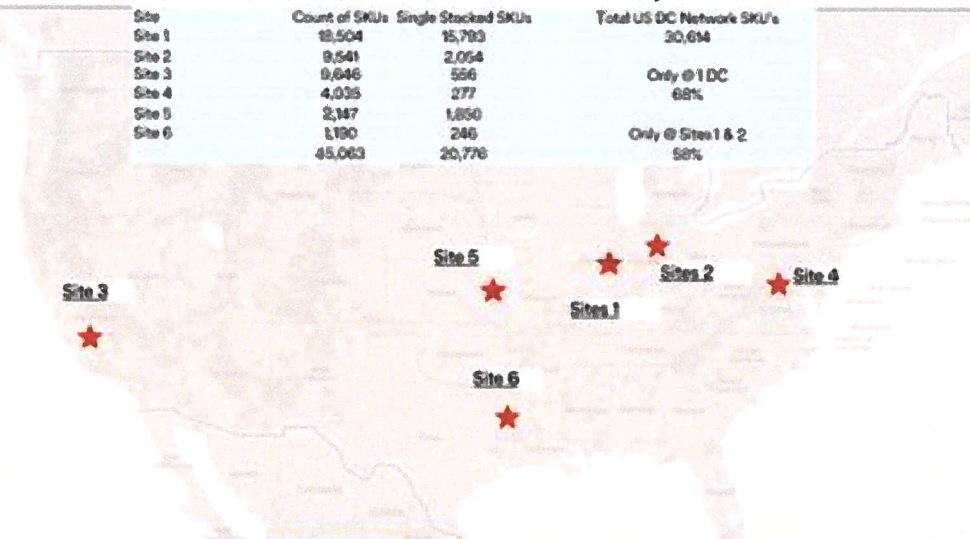


Facts / Variables

From: Current State Network - 45%

The current network achieves 45% of our volume within 1 day transit to our customer

Site	Count of SKUs	Single Stocked SKUs	Total US DC Network SKUs
Site 1	12,504	15,793	30,694
Site 2	9,541	2,054	
Site 3	9,646	556	Only @ 1 DC
Site 4	4,035	277	68%
Site 5	2,147	1,850	
Site 6	1,190	246	Only @ Sites 1 & 2
	45,063	20,776	58%



Can we meet our goal? Service, Quality & Cost

Transportation

1. Assume all inbound shipments to fulfillment centers for the future state network are coming from a centralized warehouse in St. Louis, MO. Do not evaluate the impacts of inbound shipments. That is, do not evaluate the cost of a fulfillment center's distance from St. Louis.
2. Any outbound (to customer) shipment from a fulfillment center.
 - ≤ 100 miles will cost \$0.05/lb.
 - > 100 miles to ≤ 500 miles will cost \$0.17/lb.
 - > 500 miles will cost \$0.22/lb.
3. Industry equates 500 miles to one day of trucking service. That is, for the purpose of this case, orders that must travel greater than 500 miles will not meet the 1-day transit-time requirement.



Facts / Variables

Fulfillment

- Assume production labor is net neutral. That is, do not include the cost of production labor in this analysis.
- Administrative costs of a each fulfillment center is \$1M / year.
- Space costs are \$8 / per square foot per year.
- Storage density - each square foot of floor space can hold 3.5 cubic feet of stock.
- Assume growth of demand to be 5% per year.

Inventory

Customers are partitioned into A, B and C movers based on their demand. Stocking level for each:

- A movers: 2 weeks of customer demand at each location.
- B movers: 3 weeks of customer demand at each location.
- C movers: 4 weeks of customer demand at each location.

When changing the number of fulfillment centers, assume that the new quantity of safety stock that must be maintained abides by the square root law.

Assume the carrying cost of inventory to be 20%.

The inventory value should be \$1 per billing unit (BU).

Size the fulfillment center for at least a 5-year plan.



Volume File

1	CUST_NAME	Orders	Lines	OrderBUs	STATE_ABBR	TotalCartons	TotalCube	TotalWeight	ZIP_CODE	PartNum	ItemActivityCode	PickPac
2	CUSTOMER A	33	33	9632	FL	9632	2465.792	21556.416	349814620	PART7023305	A	1Day
3	CUSTOMER A	30	30	8064	FL	8064	2064.384	18047.232	346028310	PART7023305	A	1Day
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14	CUSTOMER A	16	16	5040	FL	5040	675.36	3291.12	349814620	PART5286391	A	1Day
15	CUSTOMER A	15	15	4860	CO	4860	720	3005.93	805388958	PART7023305	A	1Day

The file provided 3 months of customer demand

Column name	Description
CUST_NAME	Coded customer name. Customers with same name are same customer.
Orders/Lines	Each order only contains one line number
OrderBUs	Billing units (the number of items in a shipment assigned their own price).
NAME/ STATE_ABBR/ ZIP_CODE	Delivery address.
TotalCartons	Cartons shipped.
TotalCube	Number of cubic feet occupied by order.
TotalWeight	Weight of order (lbs.)
ItemActivityCode	"Mover" type for customer.
PickPack	Time taken to perform PickPack for the order.



Solution / Results

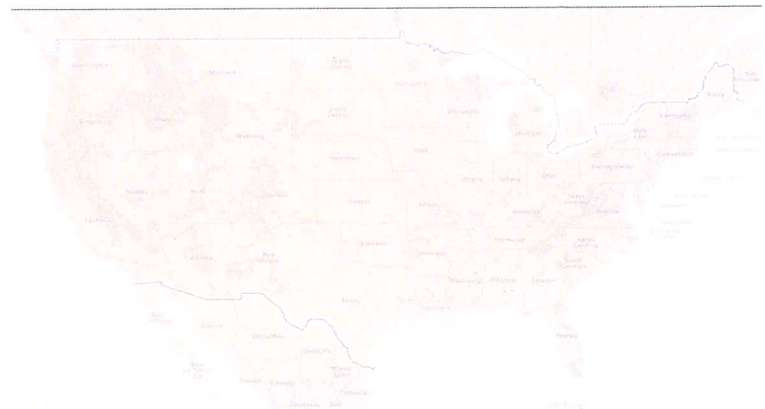
Where would you locate the Fulfillment Centers?

Proposal for an Agile Supply Chain

Recommendation - Location, Size & Throughput needs – Distribution Centers

Where would you
locate & how many
DC's do you
recommend to meet a
85 – 95% 1 day
delivery?

Make sure to balance our
inventory, operations &
trucking while meeting
service expectations.



Desired Cycle Time – 2 days (1 fulfillment & 1 transportation)

What is the square footage of each fulfillment center in your network?

What products are stocked in each site?

How much inventory of each product is stocked in each location?

What is the throughput in cartons and billing units of each site?

What is the admin cost of the proposed network?

What is the space costs of the proposed network?

What is the outbound trucking costs of the proposed network?

Time permitting: What tools & technologies do you recommend
for managing the proposed network?



Objective

The objective of supply chain network design is to balance service level against inventory carrying costs, facility costs, transportation costs and risks.

Challenge

1. Re-design the current U.S. Distribution/Transportation network for 3M, a large Multi-National Manufacturing Co. located in Minnesota.
2. Delivery should meet a 1 day transit time for 85-95% of customers
3. Counterbalances are inventory, operations & transportation costs
4. Time permitting also propose network strategies, AI tools that will enhance the network performance and management.

What stocking levels should we have?

How many sites optimize our network?

Where should we put the sites and how big will they need to be?

How should we balance redundancy against risk? (Considering weather, trucking strikes etc.)

How will the supply chain change over the next 5 years and what will that mean?

How will advances in AI and machine learning change our process?

How will we grow?
